

Demand Forecast - Center Integrated Staffing Model (ISM) Prototype

User Guide

Draft

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Prepared by

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Abstract

This document will describe the basic process flow and user interface components of the ISM Prototype. It contains four typical scenarios in using the prototype of Integrated Staffing Model to conduct staffing forecasts and miscellaneous tasks.

Version Notice

All revisions made to this document are listed below in chronological order. The release number of this document does not relate to the release number of the application:

Document Release	Date	Software Release
01	8/28/95	Prototype Draft

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1. Executive Summary

1.1 Introduction

The purpose of this document is to provide the forecasters in the Demand Forecast Center with practical guide and tutorials of using the Integrated Staffing Model Prototype.

1.2 Process and User Interface Overview

The process and user interface overviews are included with brief explanations of the functions associated with each major window of the system.

1.3 Tutorial Overview

There are four tutorials covering various aspects of the prototype. These include doing a simple forecast with ISM on the staffing level of H&PS CMC agents, using ISM forecast on number of SA Technicians and varying one external input variable, and a complex scenario of conducting an ISM forecast of POTS I&M Technicians. The last tutorial covers the use of a queuing model template independent of the ISM.

2. Introduction

2.1 Purpose

The purpose of this document is to provide the forecasters in the Demand Forecast Center with practical guide and tutorials of using the Integrated Staffing Model Prototype.

2.2 Authors

The authors of this document are Dick Hewitt, Member of Technical Staff, Modeling Group, U S WEST Advanced Technologies and Yat-Sang Hung, Member of Technical Staff, Human Factors Engineering, U S WEST Communications Information Technologies.

2.3 Project Description

The purpose for developing the ISM is to provide a forecasting method which enables the DFC to more accurately anticipate the need for U S WEST call center agents and field technicians and to strategically position U S WEST to meet and evaluate this need within an integrated framework.

The ISM incorporates a basic, high-level understanding of U S WEST operations, using the explanatory power inherent in causal models to predict the staffing impacts of future corporate decisions.

2.4 Project Scope

The prototype version of the ISM is intended to be a proof of concept. The prototype links core processes of U S WEST operations using available data and subject matter expert estimates to model the effects of actions and decisions in one core process on other processes.

The ISM prototype has been built at a high level, considering the 14 state region to be the basic unit of geography and core processes to be the basic unit of work groups.

The prototype version represents the first iteration of the the ISM System. One purpose of this prototype is to identify areas of improvement and changes for the production system.

2.5 Limitations of the ISM Prototype

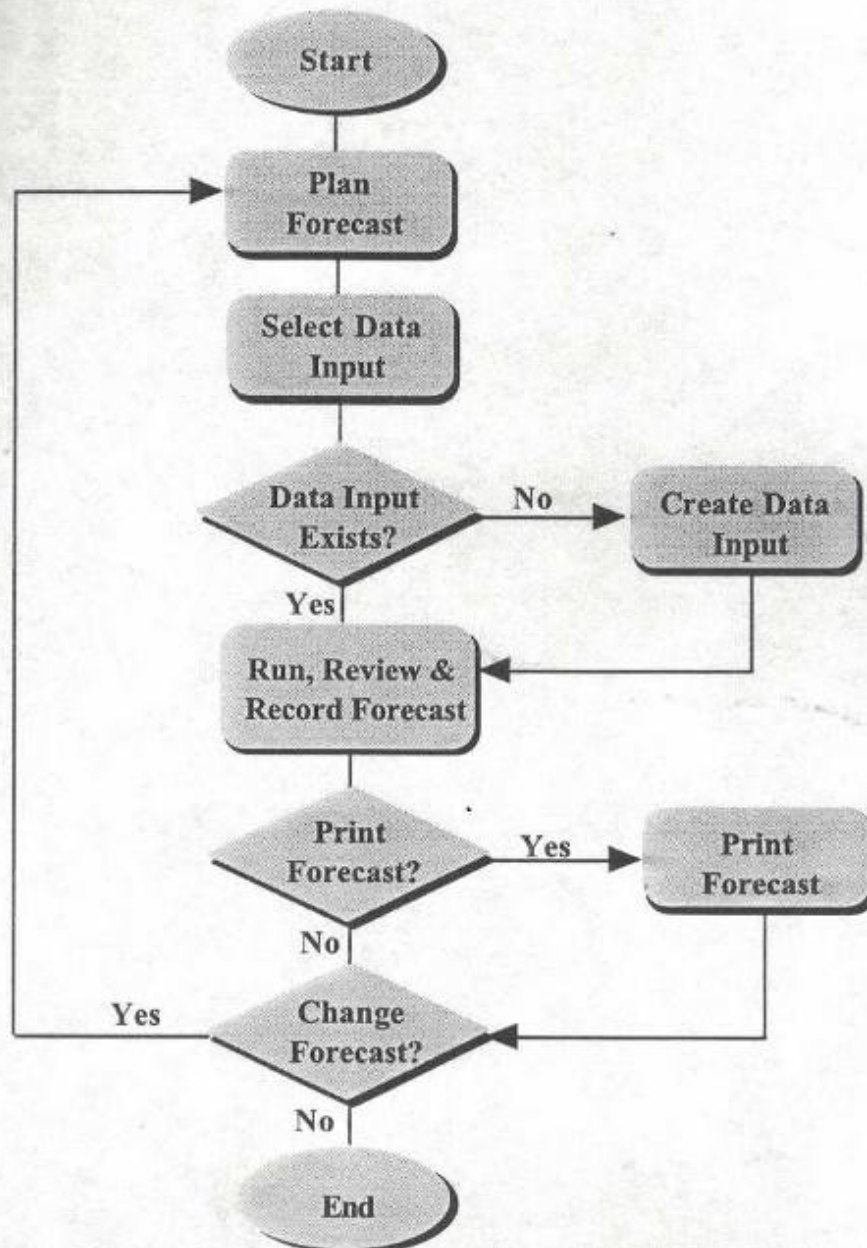
Because of the time and resource constraints in the development, the prototype is subject to the following limitations:

- The user interface design reflects the optimal use of limited resources and time rather than the optimal layout from the usability or user interface standard perspective.
- The input and output data may not be presented in a formatted and easy-to-understand form.
- The forecasts will be done at the company level.
- The data for June 95 will be available for forecasts.

3. Process Overview

This section illustrates briefly the process flow of using ISM Prototype. The details of tracing the "Influence Diagram" (Ruby Diagram) will be covered in Chapter 6, Tutorial 3 of this document.

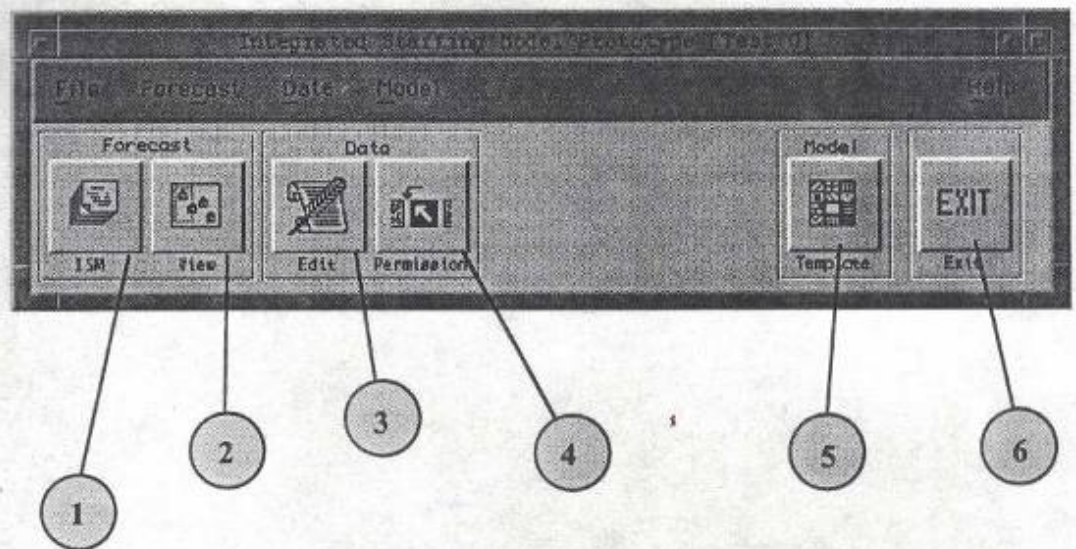
3.1 Process Flow of Using Integrated Staffing Model Prototype



4. User Interface Overview

This section describes different windows and screens of the ISM Prototype and their business functions.

4.1 ISM Prototype Main Window

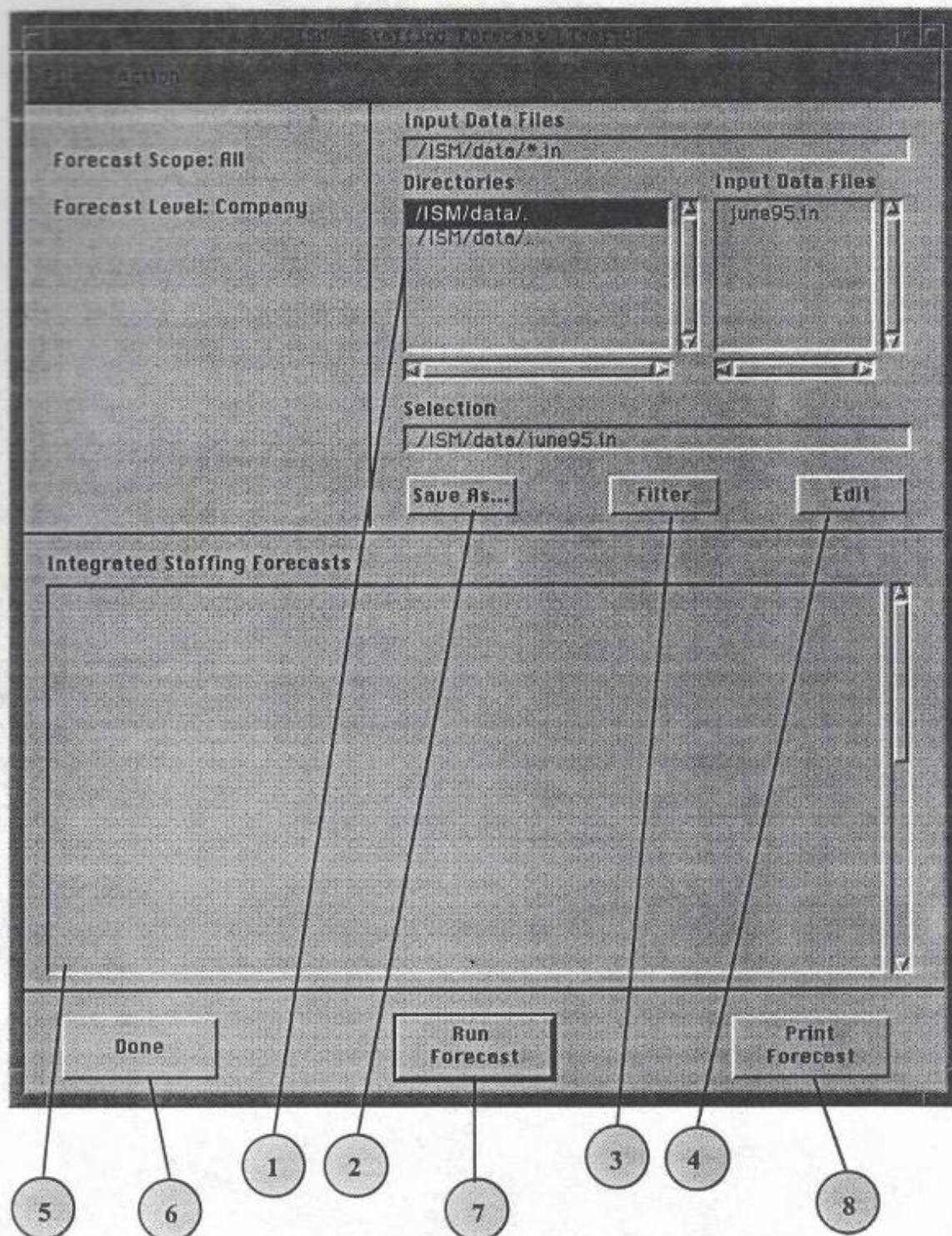


ISM Prototype Main Window

The business function of this window is to provide access to the various functions of the ISM Prototype depicted in the following:

#	Widget Name	Functions
1	Forecast ISM Iconic Button	invokes ISM Forecast Window to provide access to input data selection, copying, editing as well as running of ISM Model and the display and printing of ISM forecast results.
2	Forecast View Iconic Button	invokes ISM View Forecast Window to provide access to previously run ISM forecasts.
3	Data Edit Iconic Button	invokes Data Input File Selection Dialog Box to provide access to input data editing.
4	Data Permission Iconic Button	invokes Change Permission Dialog Box to change file permission from read-only to editable.
5	Model Template Iconic Button	invokes Model Template Window to run Queuing or Inventory Model independent of the ISM Model.
6	Exit Iconic Button	exits ISM Prototype.

4.2 ISM Staffing Forecast Window and Save As Dialog Box



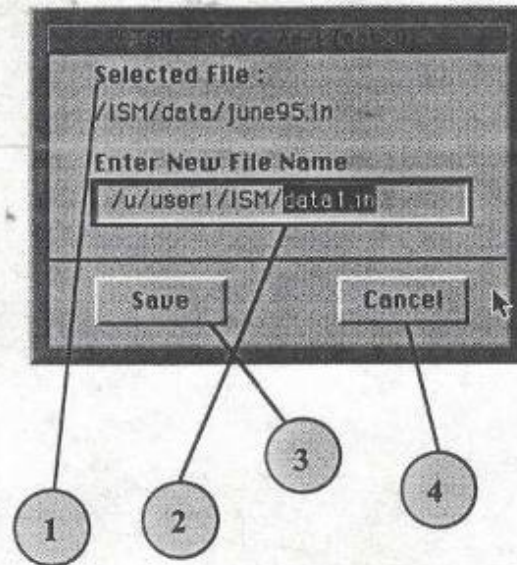
ISM - Staffing Forecast Window

The business function of this window is to provide access to input data selection, copying, editing as well as running of ISM Model and the display and printing of ISM forecast results. The functions of the labeled items are shown as follows:

#	Widget Name	Functions
1	Input Data File Selection Area	consists of <i>Data Input Files Filter</i> containing the directory mask "*.in", a <i>Directories List Box</i> , a <i>Data Input Files List Box</i> and a <i>Selection Text Field</i> . The user can navigate through the directory tree by double clicking on the corresponding item or clicking on the item and using the <i>Filter PushButton</i> (3) below.
2	Save As... PushButton	invokes <i>Save As Dialog Box</i> to copy any input file and save it as a new input file for different runs.
3	Filter PushButton	applies the directory mask to the currently selected directory.
4	Edit PushButton	invokes the default editor (as set in the environment variable of \$EDITOR of the login shell) with the selected data input file. Note that only newly created input data files (without ISM Forecast run on them) are editable. Once ISM forecast is run, the associated data input file will become read-only. If you want to edit that file, you have to change the file permission from read-only to editable.

5	Integrated Staffing Forecasts Scrollable Pane	displays the output of the ISM forecast is run. The output will be given in name-value pairs.
6	Done PushButton	closes the <i>ISM Forecast Window</i> .
7	Run Forecast PushButton	initiates calculations using the ISM Model. The results will be output to the <i>Integrated Staffing Forecasts Scrollable Pane</i> .
8	Print Forecast PushButton	prints the output forecasts and the input data file to the local printer as defined in the environmental variables \$PRINTER or \$LPDEST.

We recommend you use "Save As" to create a new input data file for each analysis you run and keep track of the changes you make for the input parameters in a spreadsheet. See Tutorial 3 for a suggested scheme. Changing a read-only data file to editable and running ISM forecast with it will overwrite the previous output forecast file.

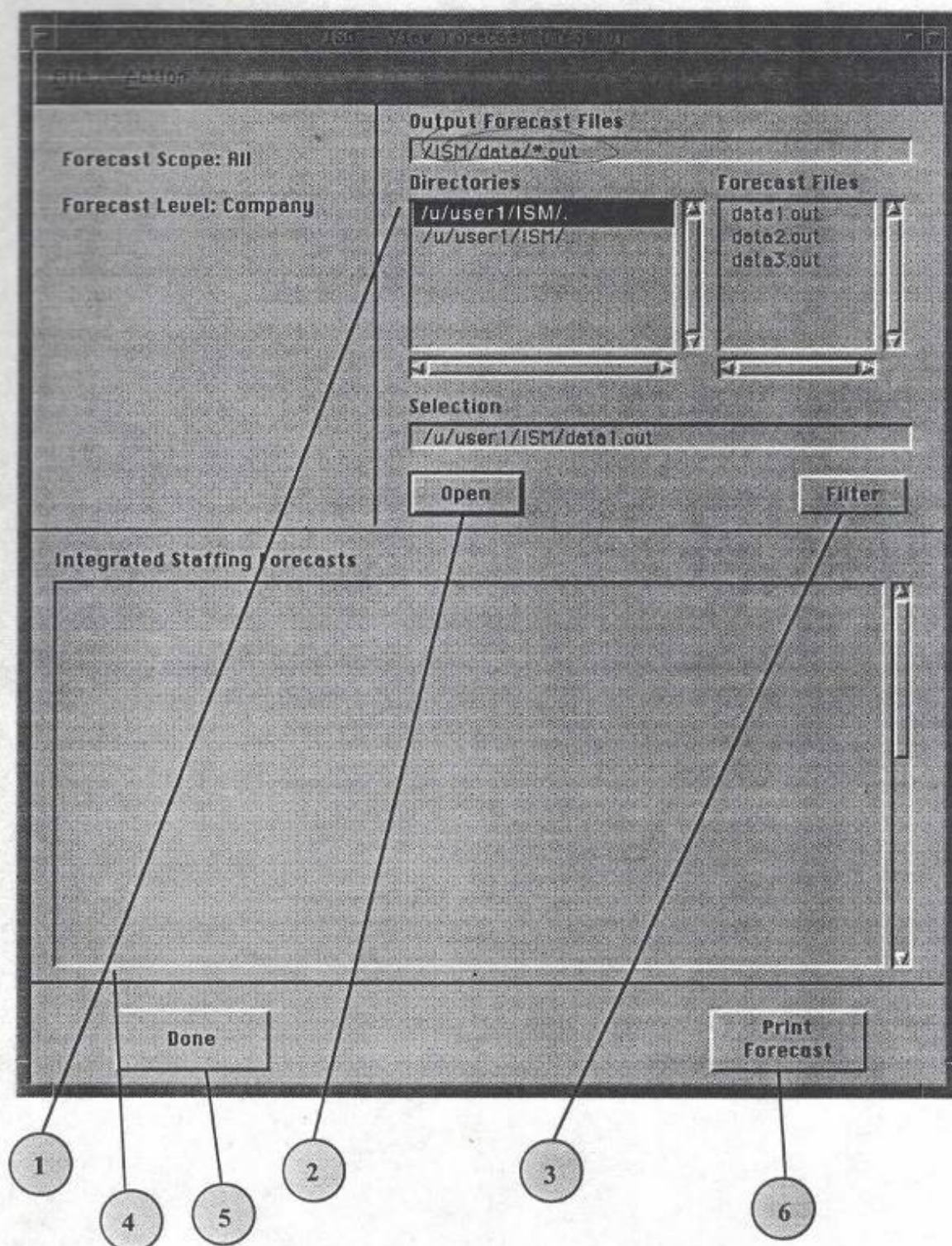


ISM - Save As Dialog Box

#	Widget Name	Functions
1	Selected File Non-Editable Text Field	displays the name of the selected file in the <i>Selection TextField</i> in the <i>ISM Forecast Window</i> .
2	Enter New File Name Text Field	allows the user to enter the name of the newly created data file (duplicate of the selected file).
3	Save PushButton	saves the new file and dismisses the dialog box.
4	Cancel PushButton	dismisses the dialog box without saving the new file.

Basic function is to duplicate data files for other runs.

4.3 ISM - View Forecast Window

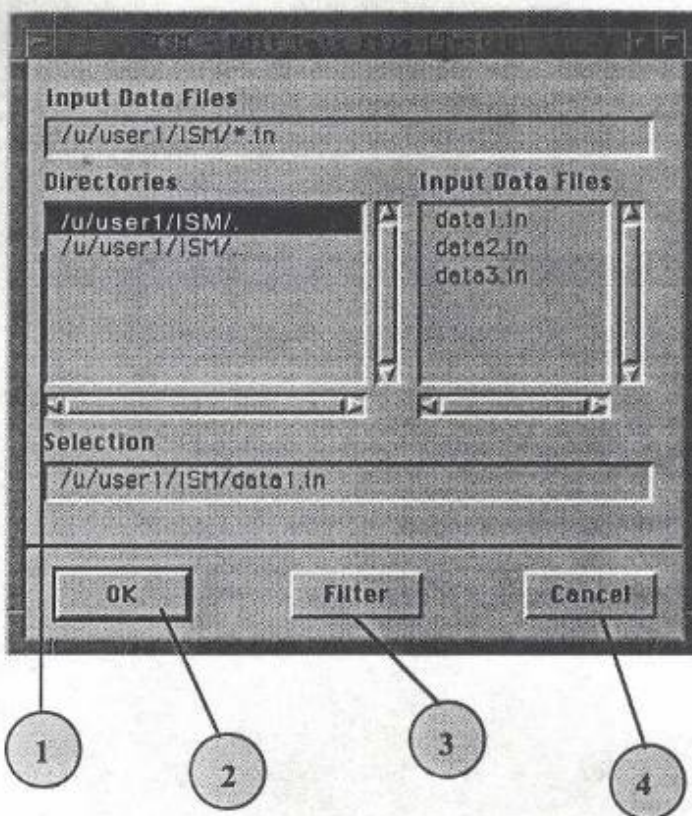


ISM - View Forecast Window

The business function of the *ISM - View Forecast Window* is to provide access to previously run output forecast data for viewing and printing. The functions of the labeled items are shown as follows:

#	Widget Name	Functions
1	Output Forecast File Selection Area	consists of <i>Output Forecast Files Filter</i> containing the directory mask "*.out", a <i>Directories List Box</i> , a <i>Forecast Files List Box</i> and a <i>Selection Text Field</i> . The user can navigate through the directory tree by double clicking on the corresponding item or clicking on the item and using the <i>Filter PushButton</i> (3) below.
2	Open PushButton	displays the contents of the output file in the <i>Integrated Staffing Forecasts Scrollable Pane</i> .
3	Filter PushButton	applies the directory mask to the currently selected directory.
4	Integrated Staffing Forecasts Scrollable Pane	displays the contents of the ISM forecast file selected. The output will be given in name-value pairs.
5	Done PushButton	closes the <i>ISM - View Forecast Window</i> .
6	Print Forecast PushButton	prints the output forecasts and the input data file to the local printer as defined in the environmental variables \$PRINTER or \$LPDEST.

4.4 ISM Data Edit File Selection Box

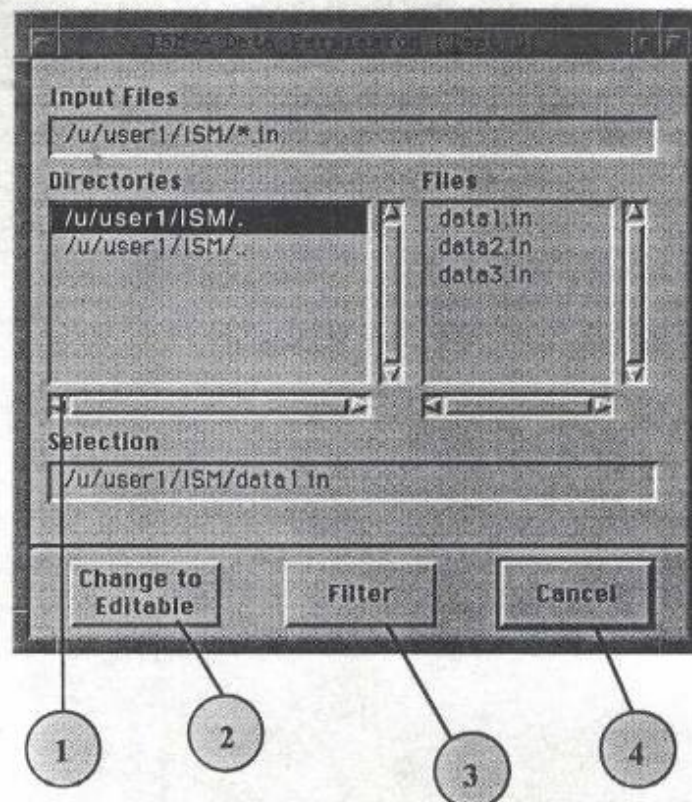


ISM - Edit Data File Selection Box

The business function of the *ISM - Edit Input Data File Selection Box* is to select editable input data file for editing. Its functions is the similar to the *Edit PushButton* in the *ISM - Staffing Forecast Window*:

#	Widget Name	Functions
1	Input Data File Selection Area	consists of <i>Data Input Files Filter</i> containing the directory mask <i>"*.in"</i> , a <i>Directories List Box</i> , a <i>Data Input Files List Box</i> and a <i>Selection Text Field</i> . The user can navigate through the directory tree by double clicking on the corresponding item or clicking on the item and using the <i>Filter PushButton</i> (3) below.
2	OK... PushButton	invokes the default editor (as set in the environment variable of <i>\$EDITOR</i> of the login shell) with the selected data input file. Note that only newly created input data files without ISM Forecast done on them are editable.
3	Filter PushButton	applies the directory mask to the currently selected directory.
4	Cancel PushButton	dismisses the file selection box without invoking the editor.

4.6 ISM - Data Permission Dialog Box

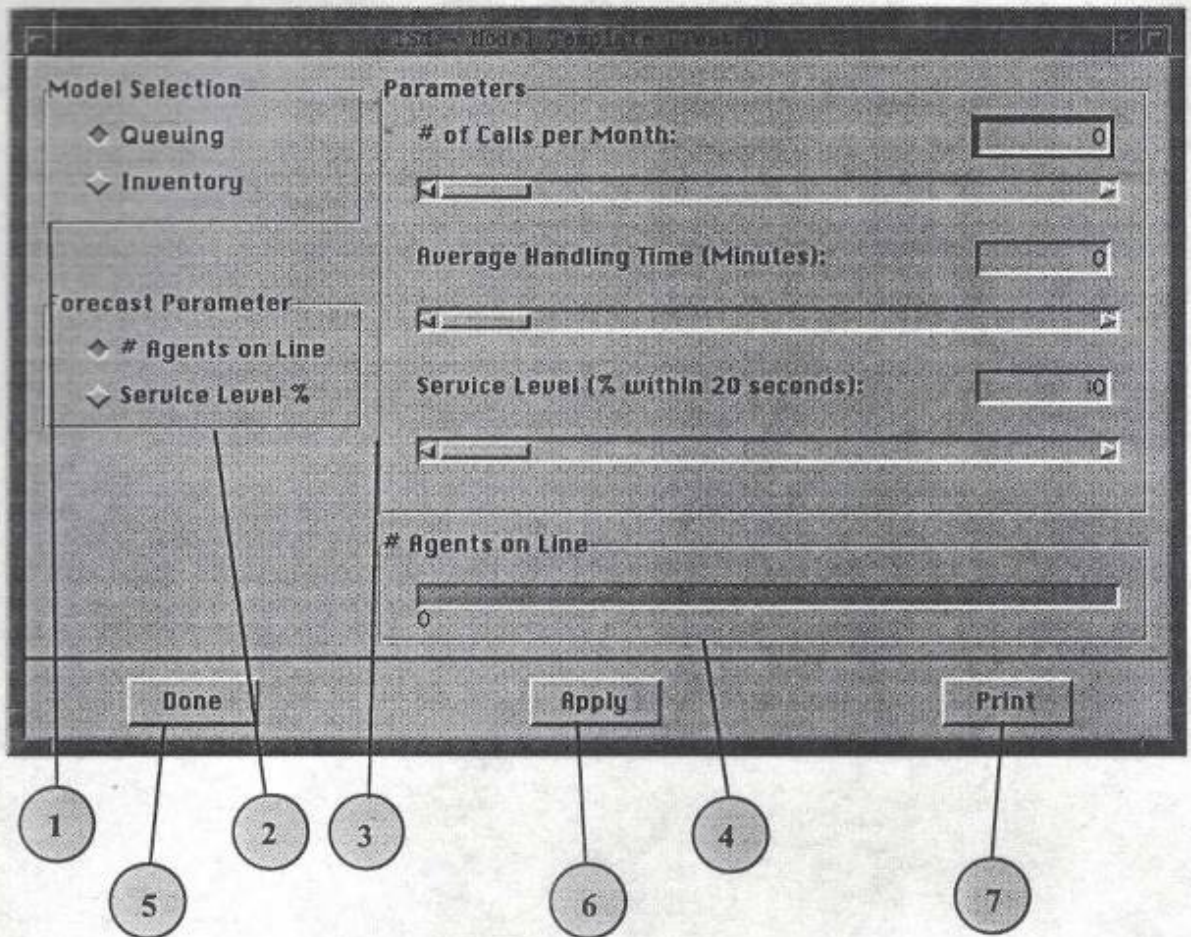


ISM - Data Permission Dialog Box

#	Widget Name	Functions
1	Data File Selection Area	consists of <i>Data Input Files Filter</i> containing the directory mask "*.in", a <i>Directories List Box</i> , a <i>Data Input Files List Box</i> and a <i>Selection Text Field</i> . The user can navigate through the directory tree by double clicking on the corresponding item or clicking on the item and using the <i>Filter PushButton</i> (3) below.
2	Change to Editable PushButton	change the permission of the selected file from read-only to editable.

3	Filter PushButton	applies the directory mask to the currently selected directory.
4	Cancel PushButton	dismisses the dialog box.

We highly recommend you
to ~~dupli~~ ~~copy~~ duplicate the file
to make changes to the
duplicate instead of using
this feature to change
the status back to ~~editable~~
read-only

4.7 ISM - Model Template ^{W.2.d.w}Dialog Box

ISM - Model Template Window : Queuing Model

ISM - Model Template Window : Inventory Model

Model Selection

- Queuing
- Inventory

Parameters

Appointments per month: 0

Average Duration of 1 Appointment (Hours): 0

Cost of Extra Technician Hour: 0

Cost of Delay per Technician Hour: 0

Technicians Hours for the month: 0

Done Apply Print

1 2 3 4 5 6 7

ISM - Model Template Window : Inventory Model

The business function of the *Model Template Window* is to enable the user to run queuing and inventory models separately without using ISM. The above example screens show the queuing model and inventory model template respectively. The functions of the labeled items are shown as follows:

#	Widget Name	Functions
1	Model Selection Area	consists of <i>Queuing Radio Button</i> and <i>Inventory Radio Button</i> . The user can select the type of model to be run by clicking on the corresponding radio button. The system default is <i>Queuing Model</i> .
2	Forecast Parameter Selection Area	consists of <i># of Agents Radio Button</i> and <i>Service Level % Radio Button</i> . This selection area is visible only when <i>Queuing Radio Button</i> in <i>Model Selection Area</i> is selected.
3	Parameters Input Area	contains the text fields and the corresponding slide bars for easy variations of the input parameters. The user can either use the slide bar or type in a value for each of the input parameters. The user clicks on <i>Apply PushButton</i> to initiate calculations.
4	Forecast Parameter Output Area	displays the output value of the forecast parameter.
5	Done PushButton	closes the <i>Model TemplateWindow</i> .
6	Apply PushButton	initiates the corresponding model calculations and displays the output in <i>Forecast Parameter Output Area</i> .

7	Print PushButton	prints the screen to the local printer as defined in the environmental variables \$PRINTER or \$LPDEST.
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When *Print PushButton* is clicked, the cursor will turn to cross-hair shape. Click anywhere on the *Model Template Window* to capture the screen and print it out.

5. Tutorial 1

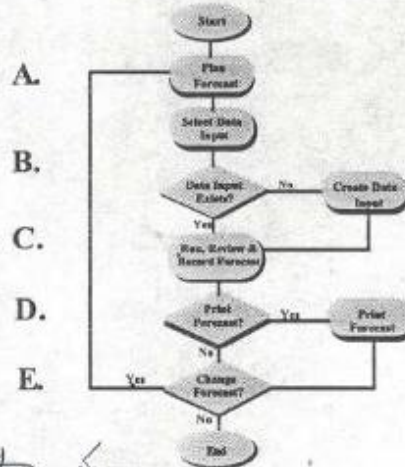
5.1 Purpose

To forecast the number of H&PS CMC Agents for June 95 using ISM and modify data to run a new forecast.

5.2 Task Overview

- Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.
- Copy the default data input file for June 95 from the system directory to your directory.
- Conduct the forecast and review the results.
- Print the forecast output and record the results
- Create a new copy of data from June 1995 and modify the copy to make a new forecast.
- Repeat steps C, D and E if necessary.

Duplicate ~~the~~
the read-only data
input file for
June 95 & save
it as an
editable file for ~~the~~ ~~forecast~~
the forecast.



5.3 Detailed Steps

- A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.

I. User Action

- a.) Locate *Group A Block A2 HPS_CMC_ENG_AGENTS* from *Group A - AGENTS* in the DFC Staffing Model Structure in the Owner's Manual.
- b.) Trace the inputs from the "Input from" column of the table. In this case *C2* is the input.
- c.) Turn to the corresponding section (i.e., *Group C - Calls, Requests and Alarms*) and trace the "Input from" column of blocks *HPS_BILL_CALLS (C2)*.
- d.) Repeat the above steps for the blocks in the "Input from" column until the "Input from" column of the block is marked with "External" or the block begins to appear as one of the traced blocks in the path (in this case, a loop occurs in the path). In this case, *HPS_BILL_CALLS (C2)* leads directly to *EX_HPS_POTS_CUST (E1)*.
- e.) If you want to visualize the influences, make a copy of the Ruby Diagram and use a highlighter to mark all the precedent blocks that provide direct and indirect inputs to the block *HPS_CMC_ENG_AGENTS*. You can also prepare a Trace Influence Spreadsheet and enter all the related parameters (see Tutorial 3 for a more complex example).

- B. Copy the default data input file for June 95 from the system directory to your directory.

1. User Action

Press *Forecast ISM Iconic Button* on the control panel of Integrated Staffing Model Prototype.



1. Click on *Forecast ISM Iconic Button*

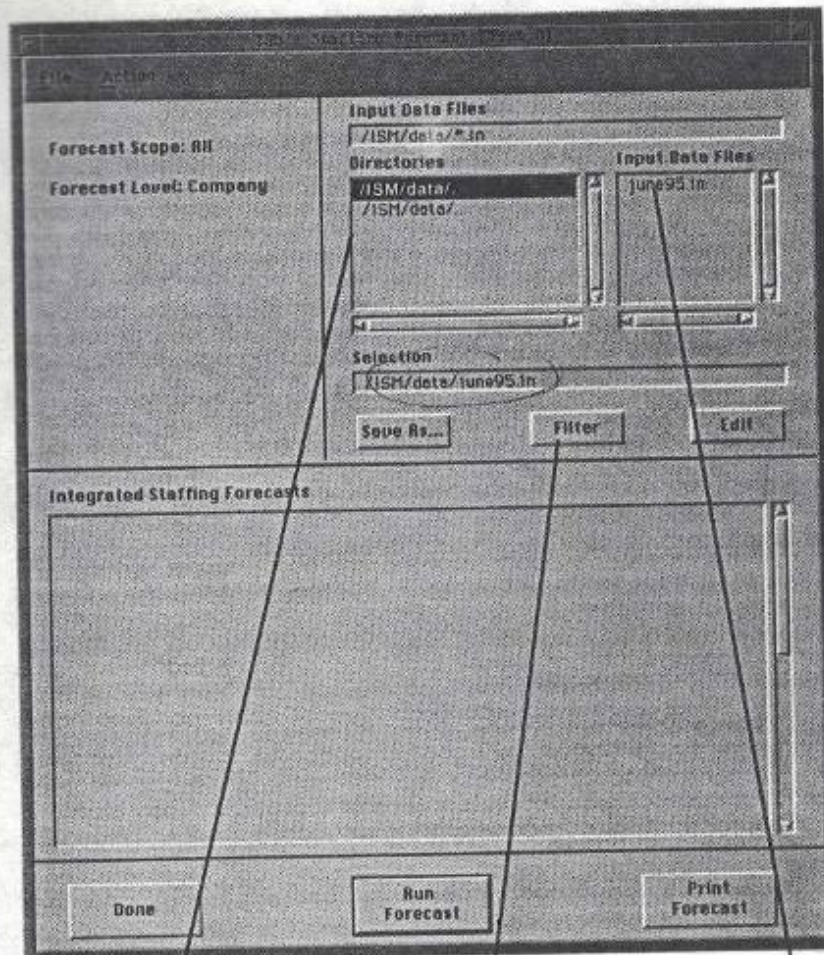
T2SMφ

1. System Response

the *ISM-Staffing Forecast Window* will appear.

- B. Copy the default data input file for June 95 from the system directory to your directory.

1. User Action *1/4/95*
- Select */ISM/data* from the Directories List Box of the ISM-Staffing Forecast Window.
 - Click on Filter PushButton to open it with the default mask */ISM/data/** in as in the Input Data Files Filter TextField. *1/4/95 1/4/95*
 - Select *june95.in* by clicking on it. The selection will be reflected in the Selection TextField.



1a. Click on the */ISM/data* or other directories to bring the path to */ISM/data* *1/20/95*
if many

1b. Click on Filter PushButton to open it with the default mask */ISM/data/** in. *1/4/95 1/4/95*

1c. Click on *june95.in* to select the file.

1.) System Response

- a.) The Selection TextField will be autopopulated to ~~/ISM/data~~ (u/ace1)
 b.) The Input Data File ListBox will display all files ending with "in".
 c.) The Selection TextField will be autopopulated to ~~/ISM/data~~ /u/user/june95.in.

/u/user/june95.in
~~at 4.4.~~

Notes:

/u/user/june95.in is the default data file for
 june 95.in It is a readonly file. ~~If you want to~~
~~edit any parameters, you~~

We recommend you to ~~use the file~~
 make ~~copies~~ of the
 copy

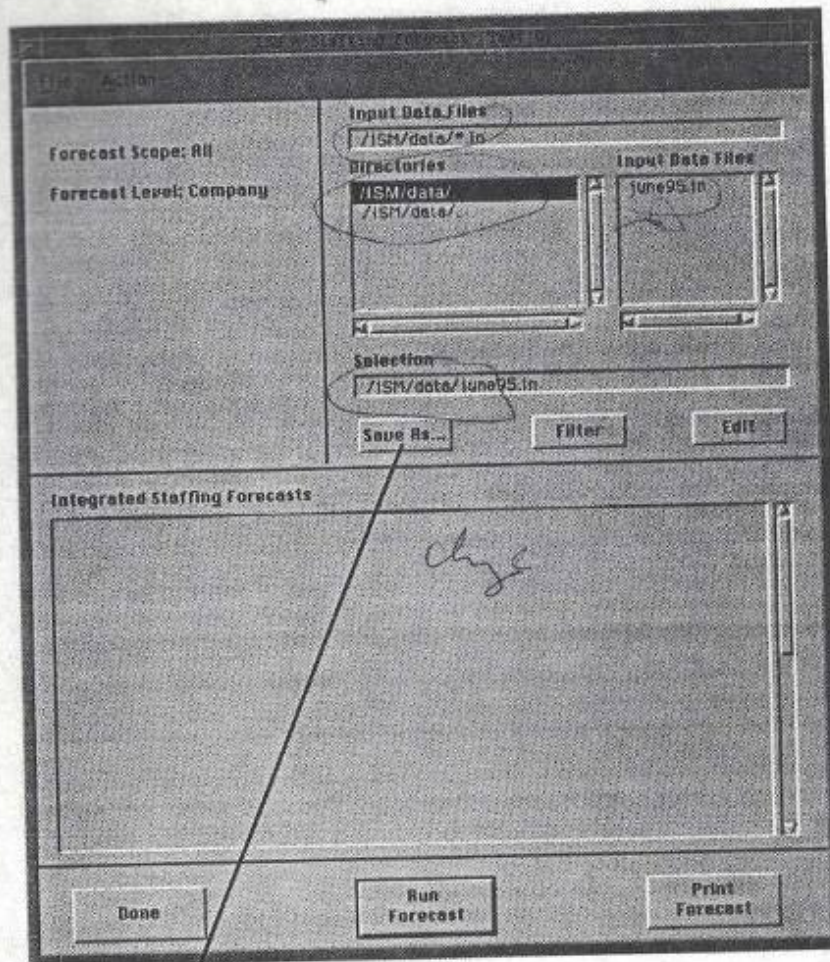
for each run you conduct.

After Foreant is run on the file, it will
 become readonly to prevent accidental change of
 input data.

- B. Copy the default data input file for June 95 from the system directory to your directory.

2. User Action

- a.) Press "Save As..." PushButton.



- 2a. Click on *Save As...* PushButton

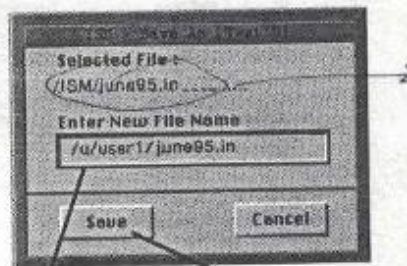
2. System Response

- a.) "Save-As" Dialog Box will appear.

- B. Copy the default data input file for June 95 from the system directory to your directory.

2. User Action

- b.) Enter the name of the newly created data file in the *Enter New File Name TextField*. Note that the complete path to your directory needs to be specified.
- c.) Press "Save" *PushButton* to save the file in your own directory. You can also modify the name to any name with "in" extension (e.g., data1.in)



2b. Enter the name of the newly created data file

2c. Click on "Save" PushButton

TZSM3

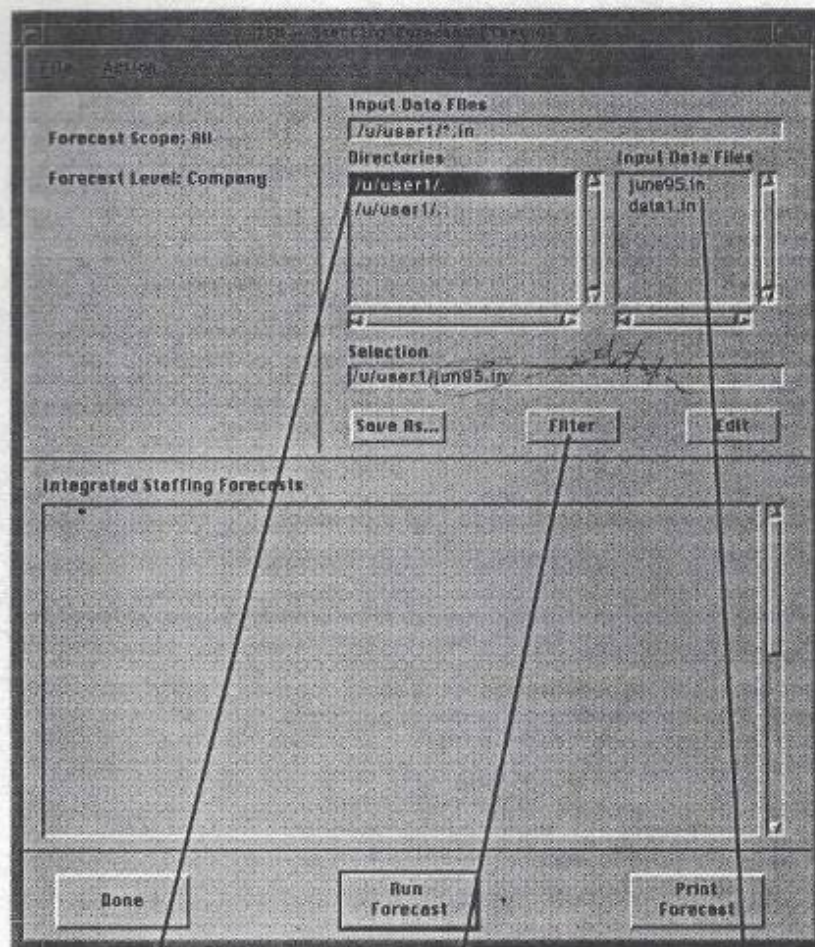
3. System Response

- c.) The file will be saved in your directory.

C. Conduct the forecast and review the results

I. User Action

- a.) Select `/u/user1` from the Directories List Box of the ISM-Staffing Forecast Window using .
- b.) Select `june95.in` from the Input Data Files List Box of the ISM-Staffing Forecast Window.
- c.) Press "Run Forecast" PushButton to initiate calculation.



1a. Click on `/u/user1` or other directories to bring the path to `/u/user1`

1b. Click on *Filter* PushButton to open it with the default mask `"*.in"`

1c. Click on *june95.in* (or other data file you create) to select the file.

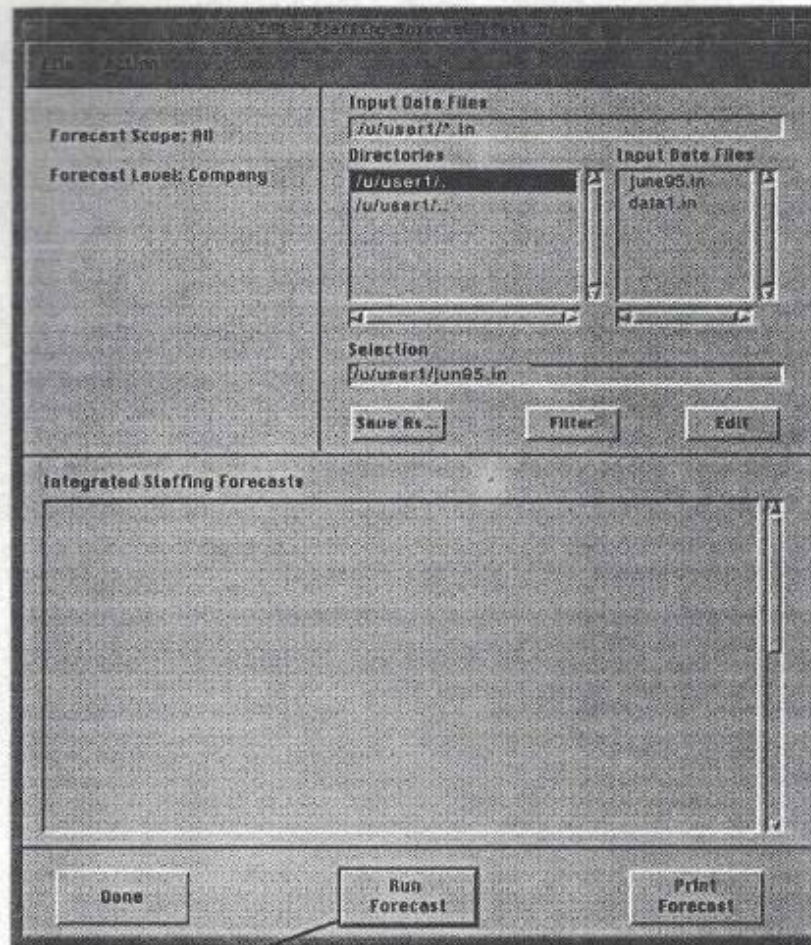
1. System Response

- a.) The *Selection TextField* will be autopopulated to /u/user1 *date/*.
- b.) The *Selection TextField* will be autopopulated to /u/user1/june95.in
- c.) The results of the calculation will be shown in the *Integrated Staffing Forecast ListBox* of the ISM-Staffing Forecast Window.

C. Conduct the forecast and review the results

2. User Action

Press "Run Forecast" PushButton to initiate calculation.



2. Click on *Run Forecast PushButton* to initiate forecast

2. System Response

The results of the calculation will be shown in the *Integrated Staffing Forecast ListBox* of the ISM-Staffing Forecast Window.

C. Conduct the forecast and review the results

3. User Action

After the forecast is done, use the slide bar of the Integrated Staffing Forecast List Box to locate the name "HPS_CMC_ENG_AGENTS" and its associated value. Review the number.

Integrated Staffing Forecasts	
ONE_CALL_CALLS	200
ALARMS	150
ALTY_MSGS	400
HPS_CTC_AGENTS	650
HPS_CMC_ENG_AGENTS	541
HPS_CMC_SPA_AGENTS	121
HPS_PMC_ENG_AGENTS	432
HPS_PMC_SPA_AGENTS	93
HPS_SMC_AGENTS	671
HPS_SBG_RSA_AGENTS	1043
SRG_CMC_AGENTS	652
SRG_SSB_AGENTS	324
SRG_CC_AGENTS	492
SRG_GSC_AGENTS	1145
SRG_BGC_AGENTS	672
SRG_CTC_AGENTS	413
SRG_OFF_AGENTS	781
BGS_BDI_AGENTS	563
BGS_SDD_AGENTS	328
BGS_SAT_AGENTS	675

3. Use slide bar to locate
"HPS_CMC_ENG_AGENTS"
Review the number.

C. Conduct the forecast and review the results

4. User Action

Since the number of existing HPS POTS customers has influence on the outputs, locate the name "EX_HPS_CUST" in the Integrated Staffing ListBox by using the slide bar. The input data is located after all the output data. Review the number. You can write down the number in the spreadsheet you prepare for run 1.

Forecast Scope: All
Forecast Level: Company

Input Data Files:
/u/user1/.in

Directories:
/u/user1/.
/u/user1/.

Input Data Files:
June95.in
data1.in

Selection:
/u/user1/jun95.in

Buttons: Save As... Filter Edit

Integrated Staffing Forecasts:

EX_HPS_POTS_CUST	450000
NEW_HPS_POTS_CUST	50000
EX_BQS_CS_CUST	9000
NEW_BQS_CS_CUST	475
EX_SBL_POTS_CUST	120000
NEW_SBL_POTS_CUST	21000
EX_BQS_CS_CUST	8000
NEW_BQS_CS_CUST	1200
EX_BQS_POTS_CUST	50000
NEW_BQS_POTS_CUST	12000
EX_CXR_CS_CUST	1300
NEW_CXR_CS_CUST	600
EX_CXR_POTS_CUST	2100
NEW_CXR_POTS_CUST	310
RELATED_WORK	10000
COIN_INS_WORK	8000
COIN_RFP_WORK	8000
COIN_COLLECT_WORK	8000
MARKET_PRED	5000
CORP_STRA_PLANS	5000

Buttons: Done Run Forecast Print Forecast

4. Use slide bar to locate "EX_HPS_POTS_CUST" Review the number.

D. Print the results

1. User Action

Print the forecast by pressing "Print Forecast" PushButton.

Input Data Files

/u/user1/jun95.in

Directories

/u/user1/

/u/user1/..

Input Data Files

June95.in

data1.in

Selection

/u/user1/jun95.in

Save As... Filter Edit

Integrated Staffing Forecasts

KZ_HRS_POTS_CUST	450000
NEW_HRS_POTS_CUST	50000
EX_SNG_DS_CUST	9000
NEW_SNG_DS_CUST	676
EX_SNG_POTS_CUST	120000
NEW_SNG_POTS_CUST	21000
EX_BGS_DS_CUST	8000
NEW_BGS_DS_CUST	1200
EX_BGS_POTS_CUST	50000
NEW_BGS_POTS_CUST	12000
EX_CXR_DS_CUST	1300
NEW_CXR_DS_CUST	600
EX_CXR_POTS_CUST	2100
NEW_CXR_POTS_CUST	310
RELATED_WORK	10000
COIN_INS_WORK	8000
COIN_REF_WORK	8000
COIN_COLLECT_WORK	5000
MARKET_FROD	6000
CORP_STRA_PLANS	5000

Done Run Forecast Print Forecast

1. Click on *Print Forecast* PushButton to print a copy.

1. System Response

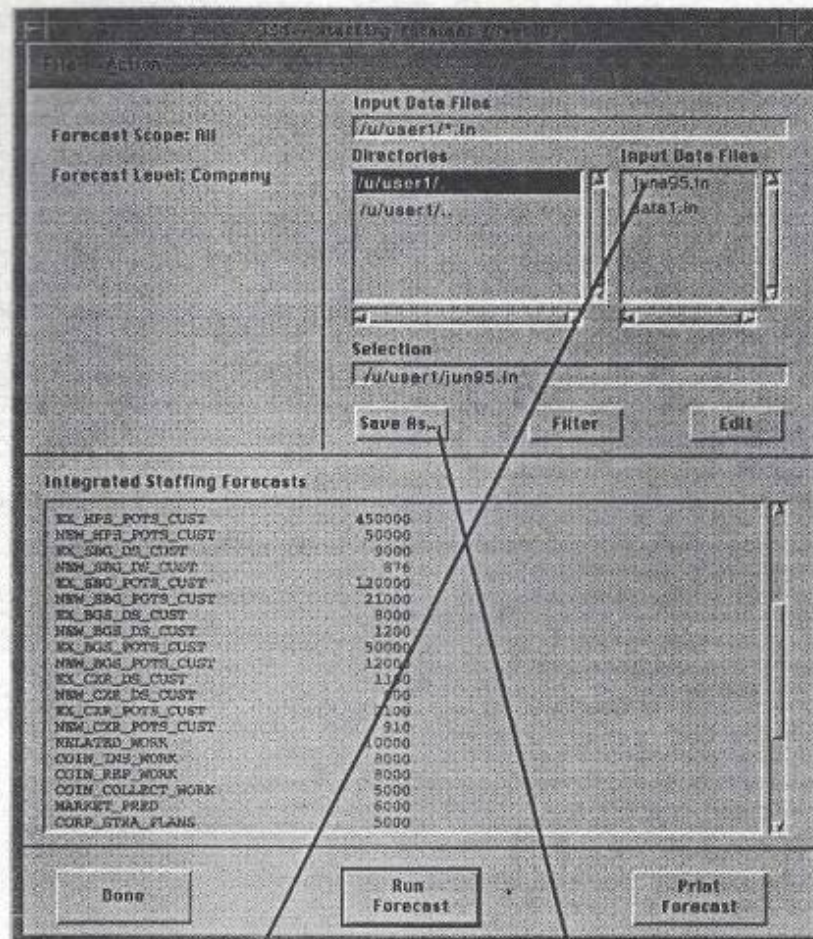
The output forecast and the input data will be printed.

- E. Create a new copy of data from ~~June 1995~~ ^{data1.in} and modify the copy to make a new forecast.

Suppose you want to reforecast June 1995 by changing the above input parameters.

1. User Action ^{data1.in}

- Select ~~June95.in~~ ^{data1.in} from the Input Data Files List Box of the ISM-Staffing Forecast Window.
- Press "Save As..." PushButton.



TZSM6

^{data1.in}
1a. Click on June95.in to select the file.

1b. Click on Save As... Button.

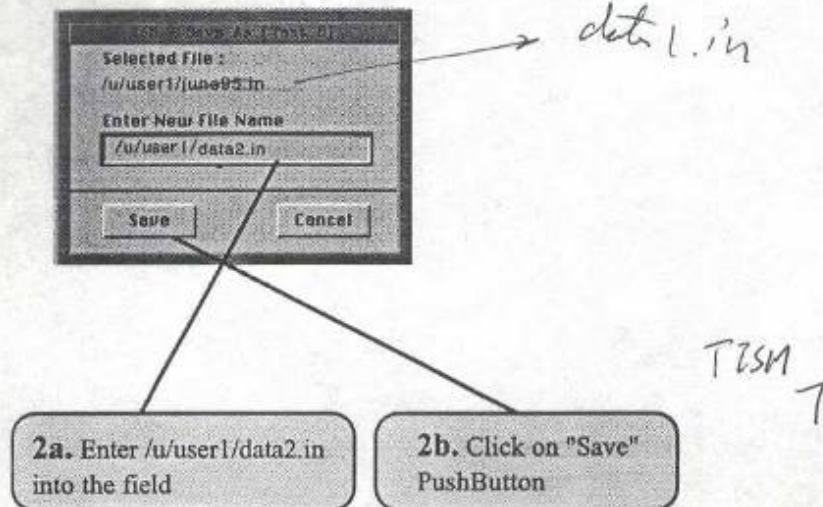
1. System Response

- a.) The Selection TextField will be autopopulated by */u/user1/june95.in*
- b.) A "Save-As" Dialog Box will appear.

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

2. User Action

- a.) Type `/u/user1/data2.in` in the *Enter New File Name TextField*.
- b.) Press "Save" *PushButton* to save the file in your own directory.

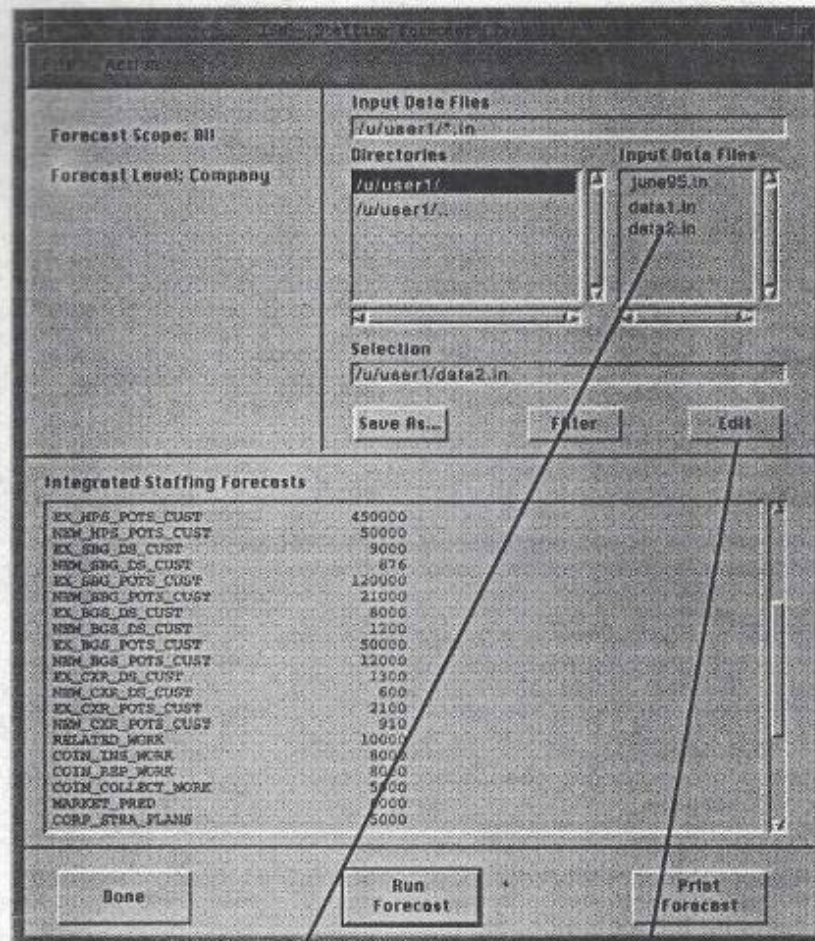
**2. System Response**

The file will be created and saved.

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

3. User Action

- a.) Select *data2.in* from the *Input Data Files List Box* of the *ISM-Staffing Forecast Window*.
- b.) Press "Edit" to invoke the default editor (e.g., vi). The editor will appear in another window.



TISM8

3a. Click on *data2.in* to select the file.

3b. Click on *Edit Button*.

3. System Response

The editor will appear in a separate window. Refer to your editor documentation for more details on how to use the editor.

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

The following actions will be different for different editors. You can change the environmental variable SEDITOR to the name of the editor you like to use.

3. User Action

- c.) Locate the input parameters you want to change.
- d.) Make the corresponding changes (e.g., locate EX_HPS_CUST and change the values associated with the name by the corresponding key strokes.).
- e.) Save the changed input data file by issuing the corresponding "Save" command of your editor.

- F. Re-run the forecast and review the results

The steps are similar to

- C. Conduct the forecast and review the results.
- D. Print the forecast output and record the results

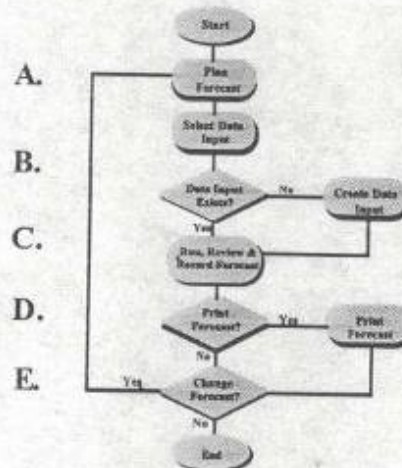
6. Tutorial 2

6.1 Purpose

To forecast the number of SA LNO Technicians for June 95 using ISM and modify data to run a new forecast.

6.2 Task Overview

- A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.
- B. Copy the default data input file for June 95 from the system directory to your directory.
- C. Conduct the forecast and review the results.
- D. Print the forecast output and record the results
- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.
- F. Repeat steps C, D and E if necessary.



6.3 Detailed Steps

A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.

1. User Action

- a.) Locate *Group T Block T6 SA_LNO_TECHS* from *Group T - LNO Technicians* in the DFC Staffing Model Structure in the Owner's Manual.
- b.) Trace the inputs from the "Input from" column of the table. In this case *W3* and *W5* are the inputs.
- c.) Turn to the corresponding section (i.e., *Group W - Works and Packages*) and trace the "Input from" column of blocks *SA_WORK (W3)* and *LOAD_DISP_WORK (W5)*. Note that *W3* also serves as an input for *W5*.
- d.) Repeat the above steps for the blocks in the "Input from" column until the "Input from" column of the block is marked with "External" or the block begins to appear as one of the traced blocks in the path (in this case, a loop occurs in the path). This step is left for the user as an exercise. For simplicity, note that one of the inputs to *W3* is *RELATED_WORK (W6)* is external input. We will change this input and look at its effect on the number of *SA_LNO_TECHS*.
- e.) If you want to visualize the influences, make a copy of the Ruby Diagram and use a highlighter to mark all the precedent blocks that provide direct and indirect inputs to the block *SA_LNO_TECHS*. You can also prepare a Trace Influence Spreadsheet and enter all the related parameters (see Tutorial 3 for an example).

- B. Copy the default data input file for June 95 from the system directory to your directory.

1. User Action

Press *Forecast ISM Iconic Button* on the control panel of Integrated Staffing Model Prototype.



1. Click on *Forecast ISM Iconic Button*

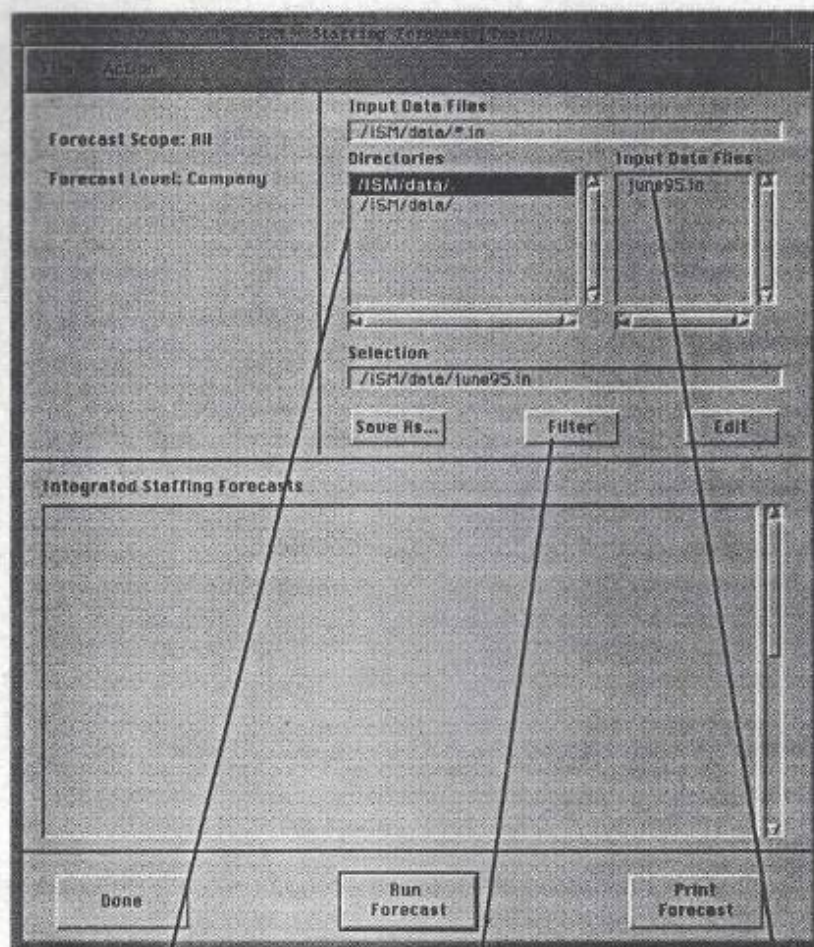
1. System Response

the *ISM-Staffing Forecast Window* will appear.

- B. Copy the default data input file for June 95 from the system directory to your directory.

1. User Action

- a.) Select `/ISM/data` from the *Directories List Box* of the *ISM-Staffing Forecast Window*.
- b.) Click on *Filter PushButton* to open it with the default mask `/ISM/data/*.in` as in the *Input Data Files Filter TextField*.
- c.) Select `june95.in` by clicking on it. The selection will be reflected in the *Selection TextField*.



1a. Click on `/ISM/data` or other directories to bring the path to `/ISM/data`

1b. Click on *Filter PushButton* to open it with the default mask `/ISM/data/*.in`.

1c. Click on `june95.in` to select the file.

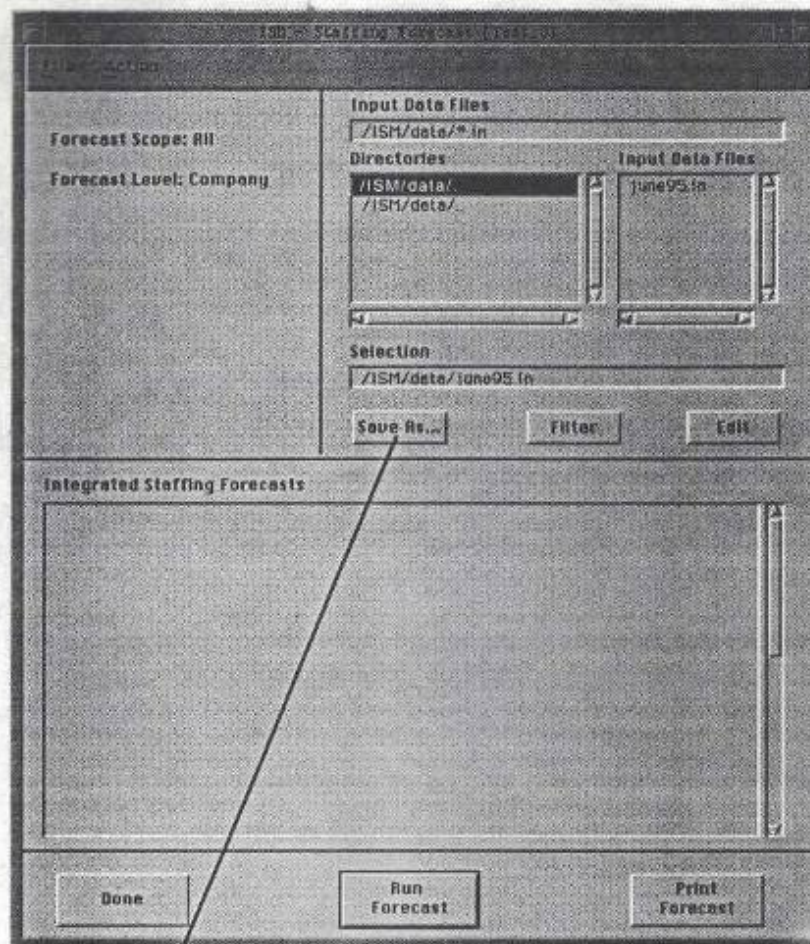
1.) System Response

- a.) The *Selection TextField* will be autopopulated to */ISM/data*
- b.) The *Input Data File ListBox* will display all files ending with "in."
- c.) The *Selection TextField* will be autopopulated to */ISM/data/june95.in.*

- B. Copy the default data input file for June 95 from the system directory to your directory.

2. User Action

- a.) Press "Save As..." *PushButton*.



2a. Click on *Save As...* *PushButton*

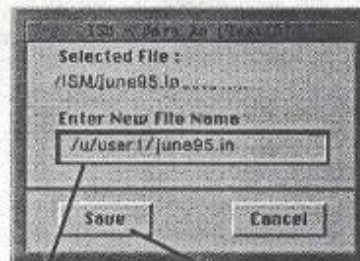
2. System Response

- a.) "Save-As" *Dialog Box* will appear.

- B. Copy the default data input file for June 95 from the system directory to your directory.

2. User Action

- b.) Enter the name of the newly created data file in the *Enter New File Name TextField*. Note that the complete path to your directory needs to be specified.
- c.) Press "*Save*" *PushButton* to save the file in your own directory. You can also modify the name to any name with "in" extension (e.g., data1.in)



2b. Enter the name of the newly created data file

2c. Click on "Save" *PushButton*

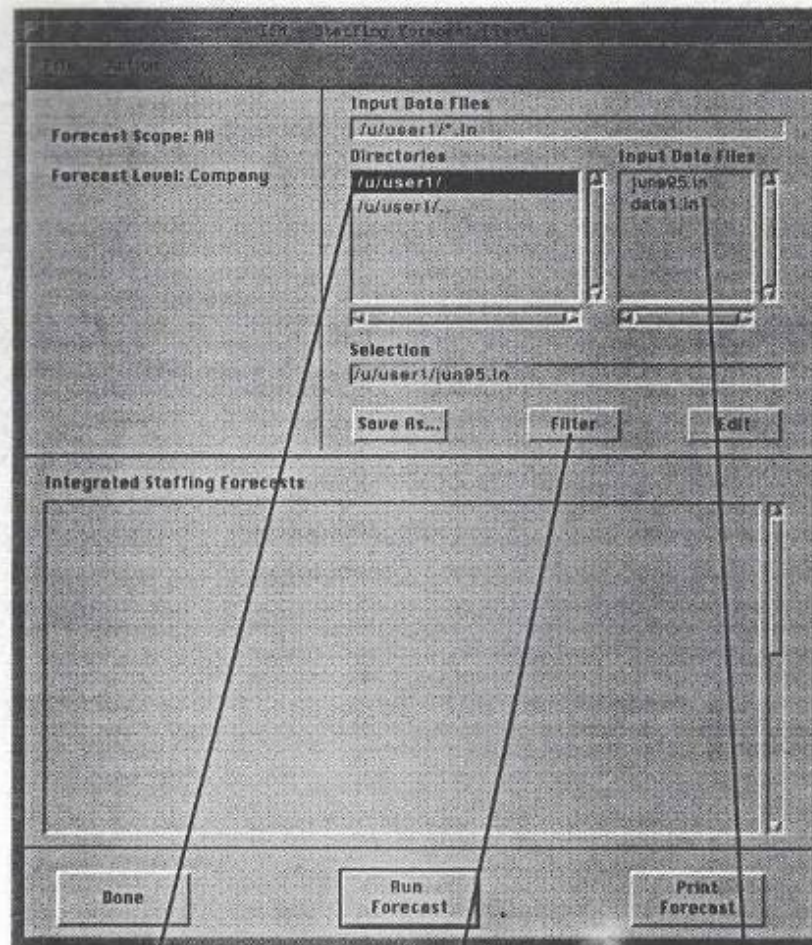
3. System Response

- c.) The file will be saved in your directory.

C. Conduct the forecast and review the results

1. User Action

- a.) Select `/u/user1` from the Directories List Box of the ISM-Staffing Forecast Window using .
- b.) Select `june95.in` from the Input Data Files List Box of the ISM-Staffing Forecast Window.
- c.) Press "Run Forecast" PushButton to initiate calculation.



1a. Click on `/u/user1` or other directories to bring the path to `/u/user1`

1b. Click on *Filter* PushButton to open it with the default mask `"*.in"`

1c. Click on *june95.in* (or other data file you create) to select the file.

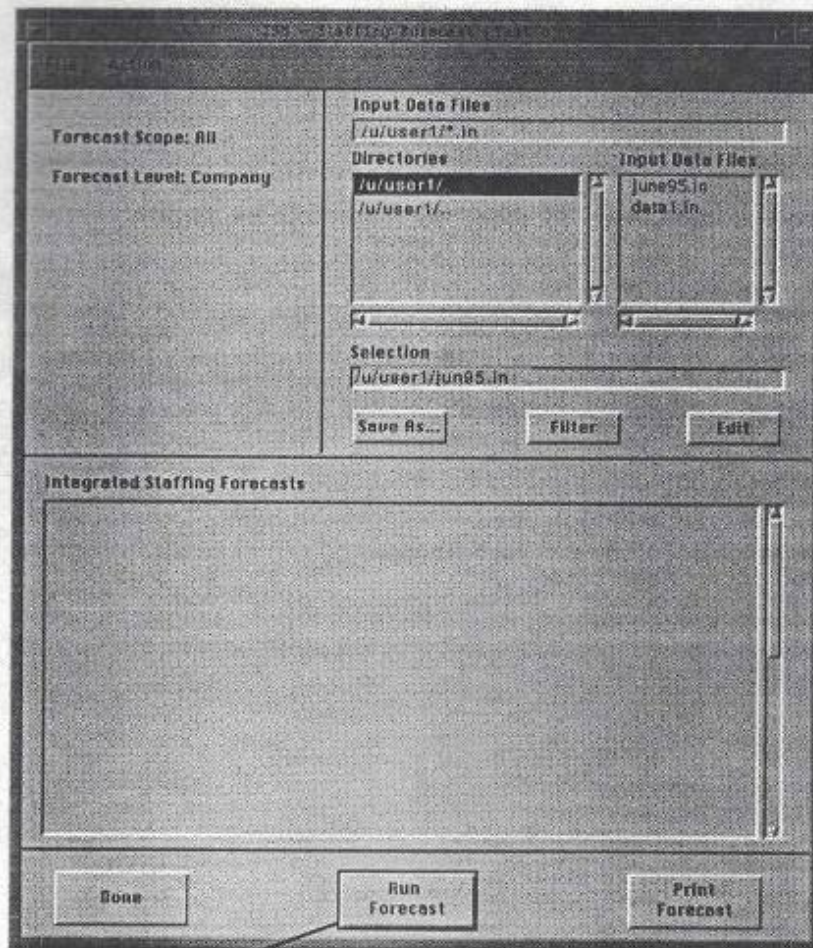
1. System Response

- a.) The *Selection TextField* will be autopopulated to */u/user1*
- b.) The *Selection TextField* will be autopopulated to */u/user1/june95.in*
- c.) The results of the calculation will be shown in the *Integrated Staffing Forecast ListBox* of the ISM-Staffing Forecast Window.

C. Conduct the forecast and review the results

2. User Action

Press "Run Forecast" PushButton to initiate calculation.



2. Click on *Run Forecast PushButton* to initiate forecast

2. System Response

The results of the calculation will be shown in the *Integrated Staffing Forecast ListBox* of the ISM-Staffing Forecast Window.

C. Conduct the forecast and review the results

3. User Action

After the forecast is done, use the slide bar of the Integrated Staffing Forecast List Box to locate the name "SA_LNO_TECHS" and its associated value. Review the number.

Forecast Scope: All
Forecast Level: Company

Input Data Files
/u/user1/*.in

Directories
/u/user1/
/u/user1/..

Input Data Files
June95.in
data1.in

Selection
/u/user1/jun95.in

Integrated Staffing Forecasts

POTS_IN_LNO_TECHS	2400
DR_DSO_LNO_TECHS	120
DR_MICAP_LNO_TECHS	60
CO_LNO_TECHS	2100
COIN_LNO_TECHS	1200
SA_LNO_TECHS	500
CONSTR_TECHS	800
CO_WORK	20000
CP_WORK	12000
SA_WORK	8000
LOAD_DISP_WORK	8000
RELATNG_WORK	8000
COIN_INS_WORK	6000
COIN_RPT_WORK	7000
COIN_COLLECT_WORK	2000
PLM_PCKGS	1000
NET_CAPAC	123000
NET_RELIAB	70
HELD_ORDERS	3000
MARKET_PRED	95000

Buttons: Done, Run Forecast, Print Forecast

3. Use slide bar to locate
"SA_LNO_TECHS"
Review the number.

SA
TECHS 2

~~10752~~

C. Conduct the forecast and review the results

4. User Action

Since the quantity of related work has influence on the outputs, locate the name "RELATED_WORK" in the Integrated Staffing Listbox by using the slide bar. The input data is located after all the output data. Review the number. You can write down the number in the spreadsheet you prepare for run 1.

Forecast Scope: All
Forecast Level: Company

Input Data Files
/u/user1/*.in
Directories
/u/user1/
/u/user1/..
Input Data Files
June95.in
data1.in
Selection
/u/user1/jun95.in
Save As... Filter Edit

Integrated Staffing Forecasts

POTS_IN_LNO_TECHS	2400
DS_DS9_LNO_TECHS	120
DS_HICAP_LNO_TECHS	88
CO_LNO_TECHS	2100
COIN_LNO_TECHS	1200
SA_LNO_TECHS	900
CONSTR_TECHS	800
CO_WORK	20000
CP_WORK	12000
SR_WORK	9000
LOAD_DISP_WORK	8000
RELATED_WORK	8000
COIN_INS_WORK	6000
COIN_RSP_WORK	3000
COIN_COLL_WGT_WORK	2000
PLM_FCSTS	1000
NET_CAPAC	123000
REP_RELIAB	70
WELD_COSTS	3000
MARKET_PEND	95000

Done Run Forecast Print Forecast

4. Use slide bar to locate
"RELATED_WORK"
Review the number.

TSAS3

D. Print the results

1. User Action

Print the forecast by pressing "Print Forecast" PushButton.

Input Data Files

Forecast Scope: All
Forecast Level: Company

Directories
/u/user1/

Input Data Files
June95.in
data1.in

Selection
/u/user1/jun95.in

Save As... Filter Edit

Integrated Staffing Forecasts

EX_HPS_POTS_CUST	450000
NEW_HPS_POTS_CUST	50000
EX_SNG_DS_CUST	9000
NEW_SNG_DS_CUST	876
EX_SNG_POTS_CUST	120000
NEW_SNG_POTS_CUST	21000
EX_BQS_DS_CUST	8000
NEW_BQS_DS_CUST	1200
EX_BQS_POTS_CUST	50000
NEW_BQS_POTS_CUST	12000
EX_CXR_DS_CUST	1300
NEW_CXR_DS_CUST	600
EX_CXR_POTS_CUST	2100
NEW_CXR_POTS_CUST	910
RELATED WORK	10000
COIN_INTL WORK	8000
COIN_RFP WORK	8000
COIN_COLLECT WORK	5000
MARKET FREQ	6000
CORP_STRAT PLANS	5000

Done Run Forecast Print Forecast

1. Click on *Print Forecast PushButton* to print a copy.

1. System Response

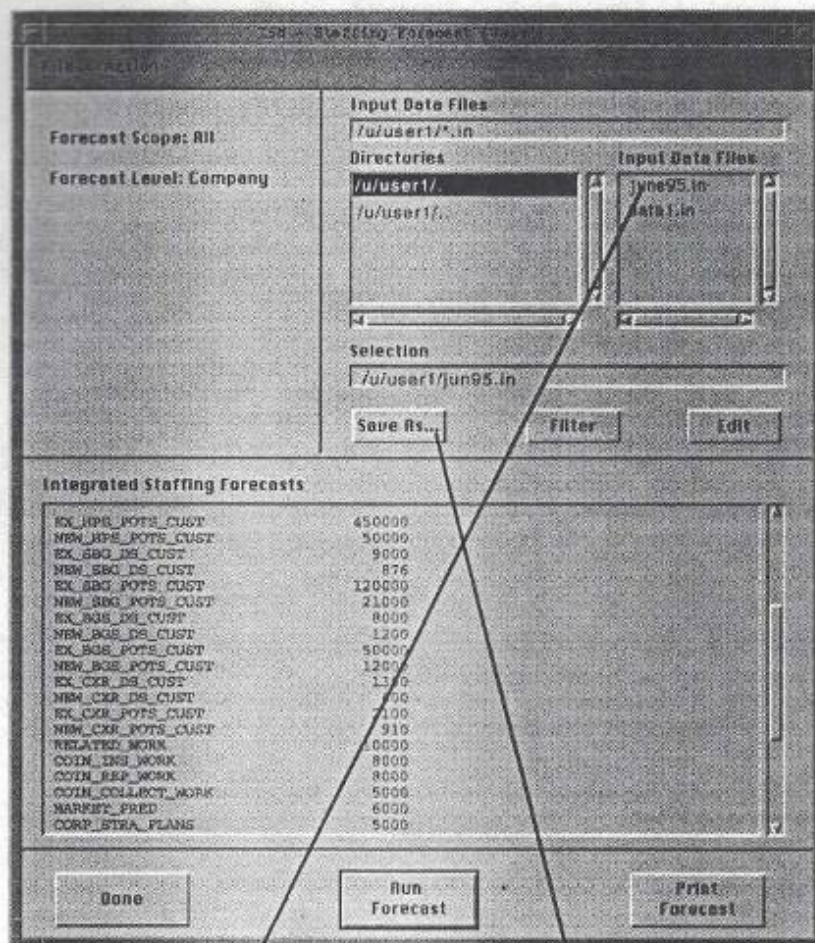
The output forecast and the input data will be printed.

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

Suppose you want to reforecast June 1995 by changing the above input parameters.

1. User Action

- Select *June95.in* from the Input Data Files List Box of the ISM-Staffing Forecast Window.
- Press "Save As..." PushButton.



1a. Click on *June95.in* to select the file.

1b. Click on *Save As...* Button.

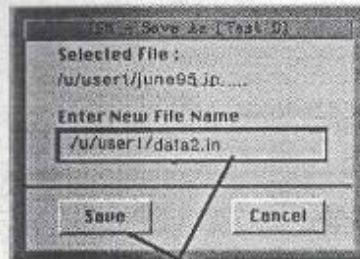
1. System Response

- a.) The Selection TextField will be autopopulated by */u/user1/june95.in*
- b.) A "Save-As" Dialog Box will appear.

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

2. User Action

- a.) Type `/u/user1/data2.in` in the *Enter New File Name TextField*.
- b.) Press "Save" *PushButton* to save the file in your own directory.



2a. Enter `/u/user1/data2.in` into the field

2b. Click on "Save" *PushButton*

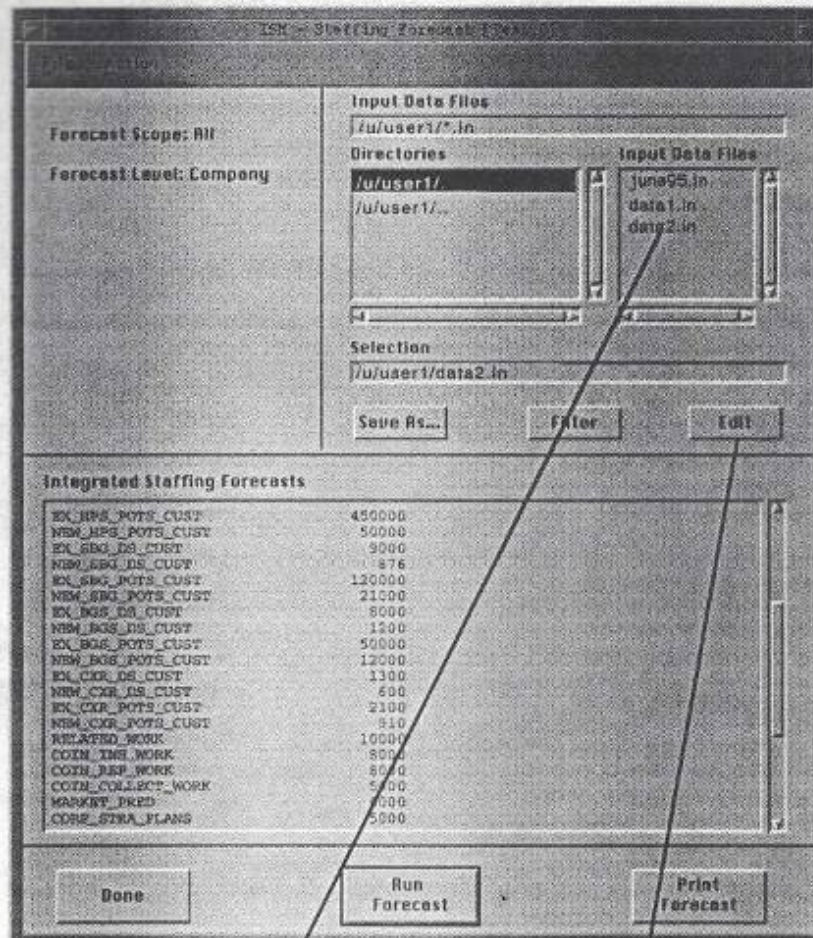
2. System Response

The file will be created and saved .

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

3. User Action

- a.) Select *data2.in* from the *Input Data Files List Box* of the *ISM-Staffing Forecast Window*.
- b.) Press "Edit" to invoke the default editor (e.g., vi). The editor will appear in another window.



3a. Click on *data2.in* to select the file.

3b. Click on *Edit Button*.

3. System Response

The editor will appear in a separate window. Refer to your editor documentation for more details on how to use the editor.

- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

The following actions will be different for different editors. You can change the environmental variable \$EDITOR to the name of the editor you like to use.

3. User Action

- c.) Locate the input parameters you want to change.
- d.) Make the corresponding changes (e.g., locate RELATED_WORK and change the values associated with the name by the corresponding key strokes.).
- e.) Save the changed input data file by issuing the corresponding "Save" command of your editor.

- F. Re-run the forecast and review the results

The steps are similar to

- C. Conduct the forecast and review the results.
- D. Print the forecast output and record the results

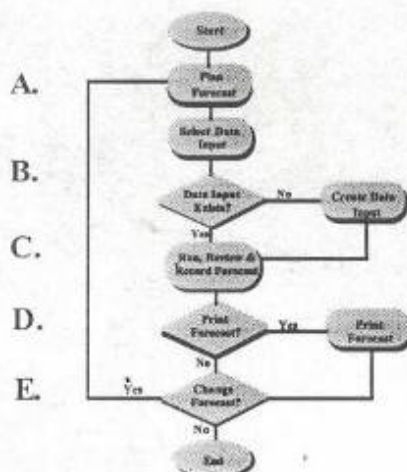
7. Tutorial 3

7.1 Purpose

To forecast the number POTS I&M LNO Technicians and explore the influences of related U S WEST processes with the Integrated Staffing Model.

7.2 Task Overview

- A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.
- B. Copy the default data input file for June 95 from the system directory to your directory.
- C. Conduct the forecast and review the results.
- D. Print the forecast output and record the results
- E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.
- F. Repeat steps C, D and E if necessary.



Detailed Steps

- A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.

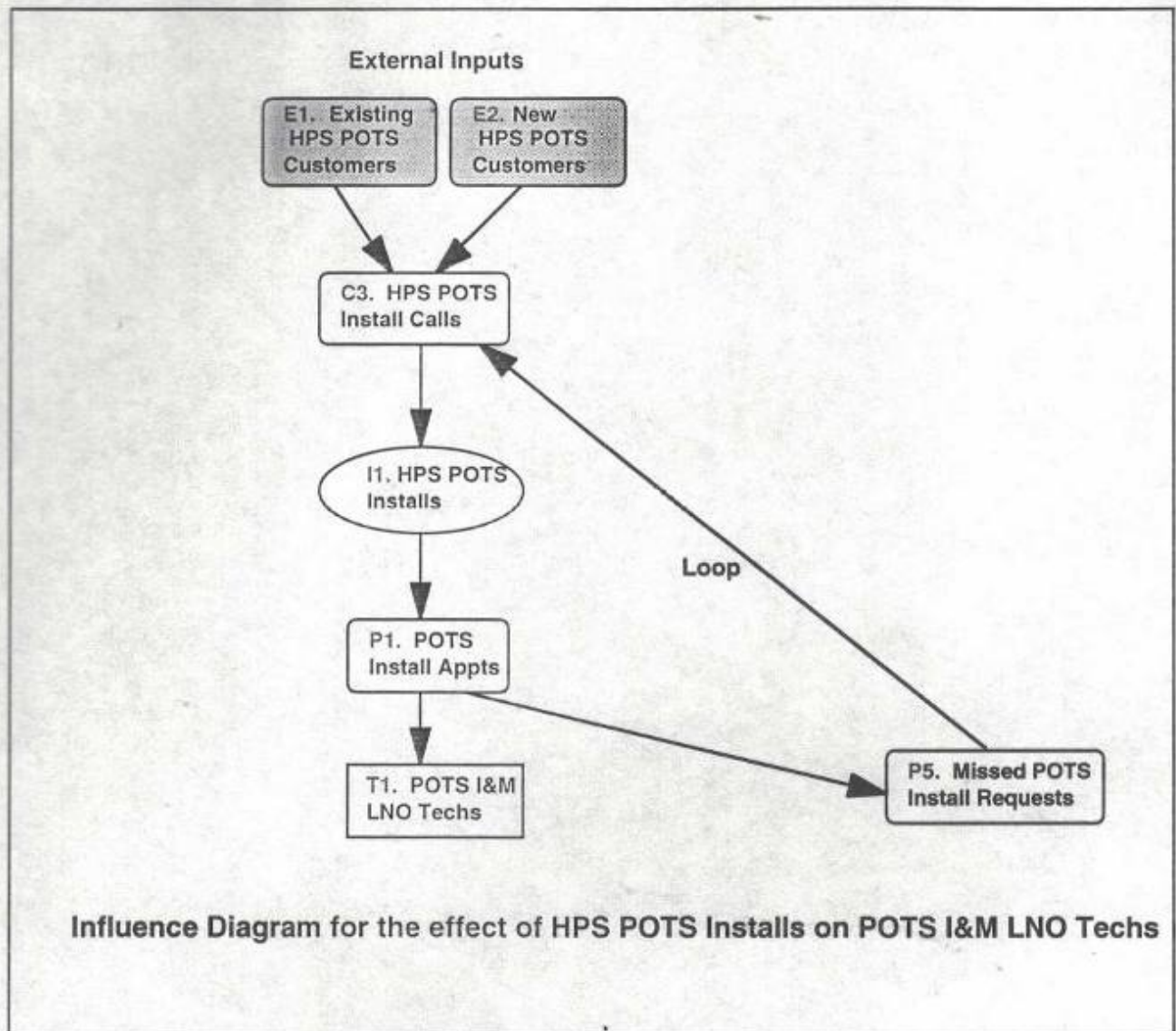
1. User Action

- a.) Locate *Group T Block T1 POTS_IM_LNO_TECHS* from *Group T - LNO Technicians* in the DFC Staffing Model Structure in the Owner's Manual.
- b.) Trace the inputs from the "Input from" column of the table. In this case *P1, P2* are the inputs.
- c.) Turn to the corresponding section (i.e., *Group P - Appointments*) and trace the "Input from" column of blocks *POTS_INS_APPTS (P1)* and *POTS_REP_APPTS (P2)*.
- d.) Repeat the above steps for the blocks in the "Input from" column until the "Input from" column of the block is marked with "External" or the block begins to appear as one of the traced blocks in the path (in this case, a loop occurs in the path).
- e.) If you want to visualize the influences, make a copy of the Ruby Diagram and use a highlighter to mark all the precedent blocks that provide direct and indirect inputs to the block *POTS I&M LNO Techs*.

Hint

The trace will become very complex quickly. You can reduce the complexity of analysis by "divide and conquer". Instead of tracing multiple input sources from a block, use a "depth-first" approach by concentrating on one input of each precedent block first (the one that is most relevant to your needs). Trace the input until it becomes a loop or external. Repeat the same process for the inputs you want to trace. You can either track the input parameters associated with each "branch" in a spreadsheet or through a visual representation of the pertinent "influence diagram" by extracting the branch out of your Ruby Diagram. The two approaches are shown below as an example.

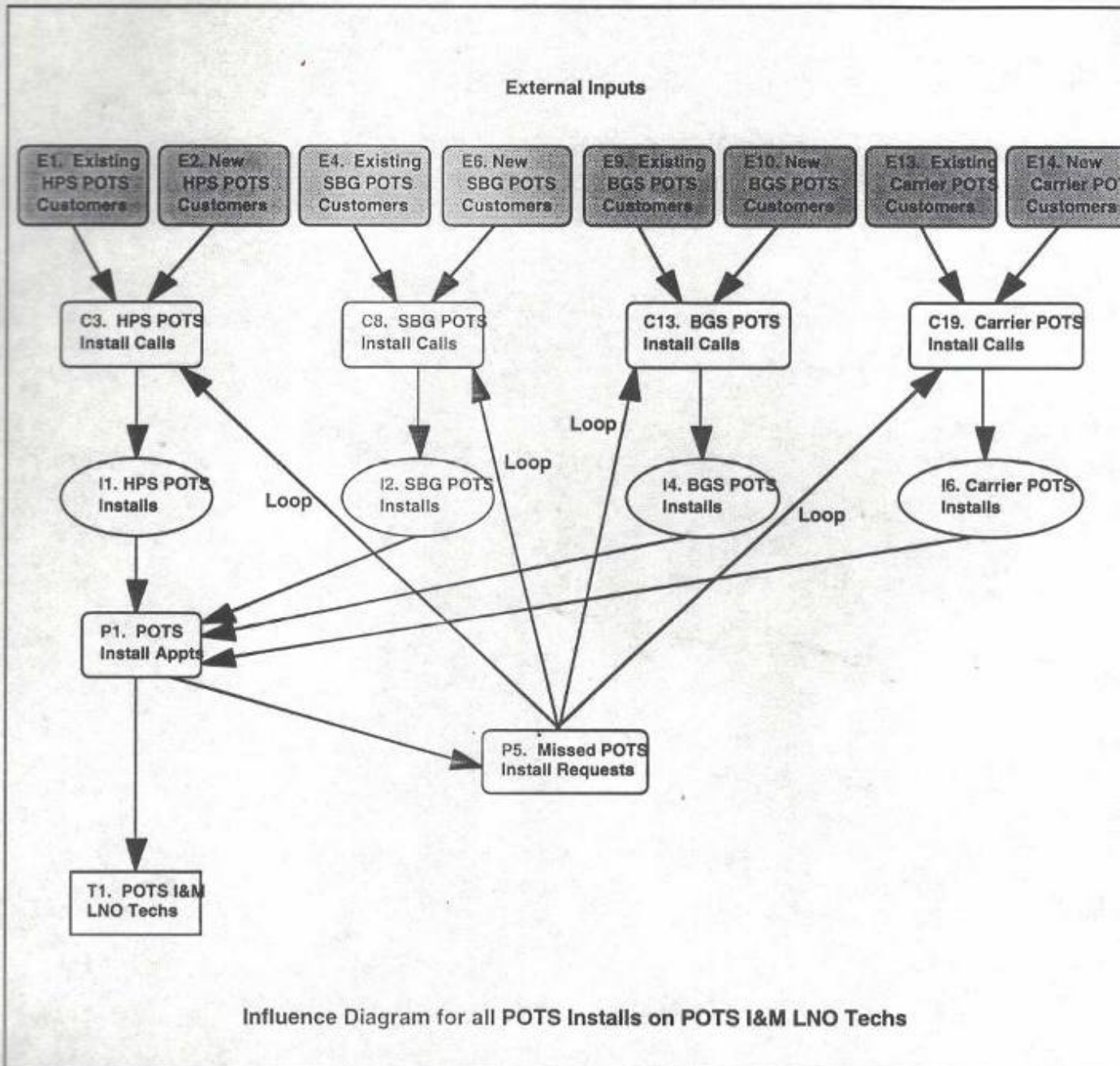
- A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.



A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.

2. User Action

- As an example, the effects of POTS Install Appts on POTS I&M LNO Technicians are analyzed using the techniques described above. The following page shows the extracted Influence Diagram.
- You can prepare a spreadsheet containing all the pertinent parameters and track their variations when you modify your external input for different runs of the forecast from the Influence Diagram.



A. Trace the influences along the path pertinent to this analysis from the ISM Ruby Diagram and the DFC Staffing Model Structure in the Owner's Manual.

Forecast Output	Run 1	Run 2	Run 3	Run 4
T1. POTS I&M Techs				
Calculated Quantities				
P1. POTS Install Appts				
I1. HPS POTS Installs				
I2. SBG POTS Installs				
I4. BGS POTS Installs				
I6. Carrier POTS Installs				
C3. HPS POTS Install Calls				
C8. SBG POTS Install Calls				
C13. BGS Install Requests				
C19. Carrier Install Requests				
P5. Missed POTS Install Requests				
External Inputs				
E1. Existing HPS POTS Customers				
E2. New HPS POTS Customers				
E4. Existing SBG POTS Customers				
E6. New SBG POTS Customers				
E8. Existing BGS POTS Customer				
E10. New BGS POTS Customers				
E13. Existing Carrier POTS Customers				
E14. New Carrier POTS Customers				

The Trace Influence Spreadsheet

- B. Copy the default data input file for June 95 from the system directory to your directory.

Part B is the same as Tutorial 1 and 2. Please refer to those tutorials

C. Conduct the forecast and review the results

Part C Step 1 and 2 are the same as in Tutorial 1 and 2. Please refer to those tutorials

3. User Action

- a.) After the forecast is done, use the slide bar of the Integrated Staffing Forecast List Box to identify *POTS_IM_LNO_TECHS*. Review the number and enter it into the spreadsheet.

Integrated Staffing Forecasts	
POTS_IM_LNO_TECHS	2400
DIS_DISP_LNO_TECHS	120
DIS_MTCAP_LNO_TECHS	68
CO_LNO_TECHS	2100
COIN_LNO_TECHS	1200
SA_LNO_TECHS	900
CONSTR_TECHS	800
CO_WORK	20000
CF_WORK	12000
SA_WORK	9000
LOAD_DISP_WORK	8000
RELATED_WORK	8000
COIN_TMS_WORK	6000
COIN_REP_WORK	3000
COIN_COLLECT_WORK	2000
PIM_POWER	1000
NMT_CAPAC	123000
NMT_RELAYS	75
USED_ORDERS	3000
MARKET_PRED	95000

3. Use slide bar to locate
"POTS_IM_LNO_TECHS"
Review the number.

~~POTS2~~

POTS2

C. Conduct the forecast and review the results

4. User Action

- b.) Identify all other parameters as in the spreadsheet you prepare by using the slide bar. Review all the numbers and enter them into the spreadsheet under "Run 1".

ISM - Staffing Forecast

Forecast Scope: All
Forecast Level: Company

Input Data Files
/u/user1/*.*.in

Directories
/u/user1/

Input Data Files
June95.in
date1.in

Selection
/u/user1/jun95.in

Save As... Filter Edit

Integrated Staffing Forecasts

POTS_LMO_TECHS	2400
IS_DSQ_LMO_TECHS	120
IS_HICAP_LMO_TECHS	68
CO_LMO_TECHS	2100
COIN_LMO_TECHS	1200
SA_LMO_TECHS	900
CONSTR_TECHS	800
CO_WORK	20000
CP_WORK	12000
SA_WORK	9000
LOAD_DISP_WORK	8000
RELATED_WORK	8000
COIN_INS_WORK	6000
COIN_REP_WORK	3000
COIN_COLLECT_WORK	2000
PLM_FCKGS	1000
NET_CAPAC	123000
NET_RELIAS	79
HELD_ORDERS	3000
MARKET_FRED	95000

Done Run Forecast Print Forecast

3. Use slide bar to locate other parameters in the spreadsheet. Review the numbers.

TPOTS3

D. Print the results

Part D is the same as Tutorial 1 and 2. Please refer to those tutorials

E. Create a new copy of data from June 1995 and modify the copy to make a new forecast.

Part E Steps 1,2 and 3a and 3b are the same as those in Tutorial 1 and 2. Please refer to those tutorials.

The following actions will be different for different editors. You can change the environmental variable \$EDITOR to the name of the editor you like to use.

3. User Action

- c.) Locate the input parameters you want to change.
- d.) Make the corresponding changes (e.g., locate EX_HPS_CUST or EX_CXR_POTS_CUST and change the values associated with the name by the corresponding key strokes.).
- e.) Save the changed input data file by issuing the corresponding "Save" command of your editor.

F. Re-run the forecast and review the results

The steps are similar to

C. Conduct the forecast and review the results.

D. Print the forecast output and record the results

The analysis of the impact of POTS Repair Appts on POTS I&M LNO Techs (which is a more complicated scenario) will be left as an exercise for the users.

8. Tutorial 4

8.1 Purpose

To use generic model template and forecast number of agents in a queuing model without the Integrated Staffing Model

8.2 Task Overview

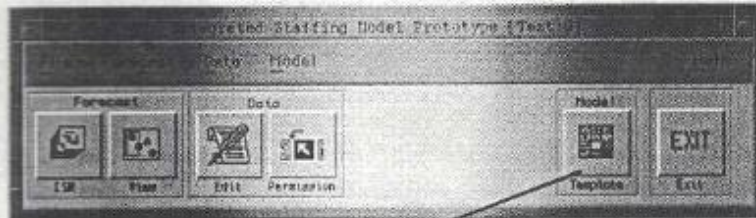
- A. Invoke the generic model template module for the queuing model.
- B. Select the number of agents as forecast parameter.
- C. Enter values for the other input parameters and conduct the forecast.
- D. Review the results and repeat step C if necessary.

Detailed Steps

A. Invoke the generic model template module for the queuing model.

1. User Action

Press "*Model Template*" Iconic Button on the control panel of Integrated Staffing Model Prototype.



1. Click on *Model Template* Iconic Button

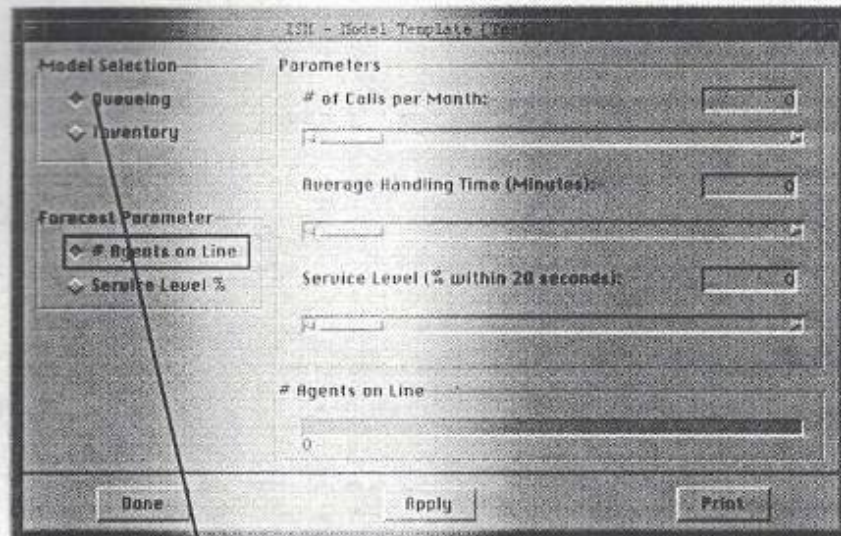
1. System Response

the *Model Template Window* will appear.

A. Invoke the generic model template module for the queuing model.

2. User Action

- a.) Select *Queuing Radio Button* from Model Selection Pane of the Model Template Window.



2. Click on *Queuing
RadioButton* initiate

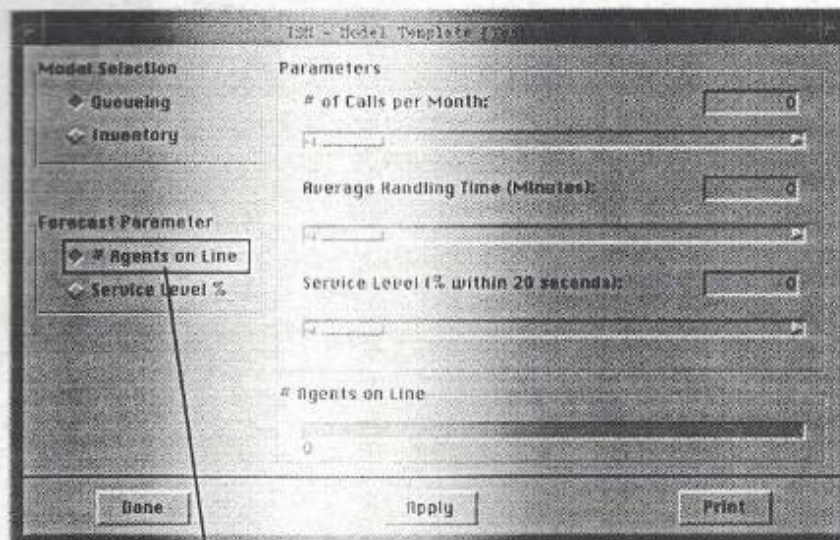
2.) System Response

- a.) The Forecasted Parameter Pane will change to reflect the appropriate model (there will not be any choice of forecasted parameter if Inventory Model is selected)

B. Select the number of agents as forecast parameter.

1. User Action

Select *# of Agents on Line* Radio Button from the Forecasted Parameter Pane of the Model Template Window.



1. Click on *# Agents on Line* RadioButton

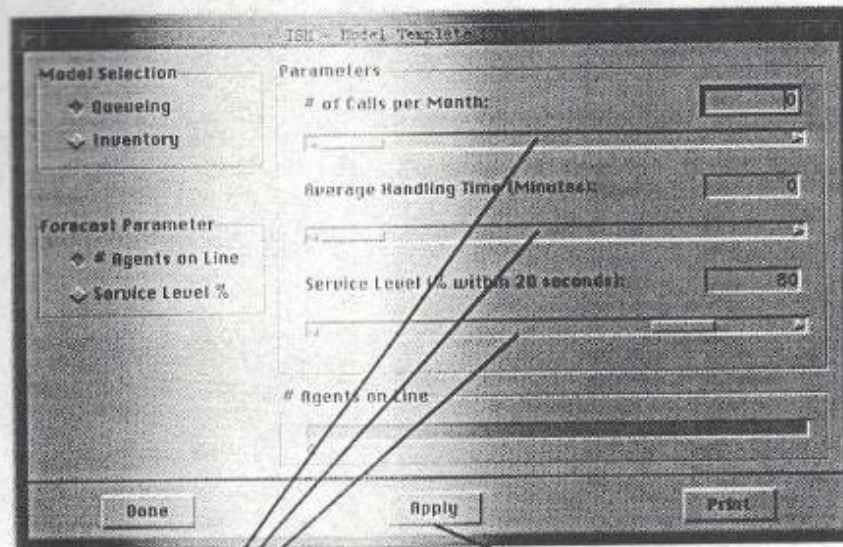
1. System Response

The window will change to the appropriate layout for the forecast parameter as shown

C. Enter values for the other input parameters and conduct the forecast.

1. User Action

- Enter values of the parameters by the corresponding slide bar or textFields.
- Press "Apply" PushButton to initiate calculations.



1a. Enter parameter values through slidebars or textFields

1b. Click on "Apply" PushButton to initiate

1. System Response

- The sidebar and the textField for each parameter will be updated accordingly.
- The result (read-only) will be shown in the # of Agents slide bar with the forecasted number.

D. Review the results and repeat Step C if necessary.

1. User Action

- a.) Review the output (# of Agents on Line)
- b.) Repeat the above step if necessary.

ISM - Model Template [1]

Model Selection

- ☒ Queueing
- ☐ Inventory

Forecast Parameter

- ☒ # Agents on Line
- ☐ Service Level %

Parameters

of Calls per Month: 100000

Average Handling Time (Minutes): 10

Service Level (% within 20 seconds): 80

Agents on Line: 541

Buttons: Done, Apply, Print

1. Review the number and repeat step C if necessary.

